# Chapter 5: Integrity Constraints

# Content

1. Introduction to Integrity Constraints (IC)
2. Characteristics of an Integrity Constraint
3. Classification of Integrity Constraints
4. Summary Impact Analysis Table

# 1. Introduction

- Integrity constraints are rules or conditions derived from the real-world requirements of the application, and these conditions are mandatory.

  ⇒ Therefore, the database must **always** satisfy all integrity constraints after any operation that modifies its data.

# 2. Characteristics of an IC

2.1 Content

2.2 Context

2.3 Impact Analysis Table

# 2.1 Content

- The content is a precise description of the meaning of the integrity constraint.

- It can be expressed in:
  - **Natural language**: easy to understand, but not always logically rigorous.
  - **Formal language** (predicate logic, relational algebra, pseudocode, etc.): concise and logically rigorous.

# 2.2 Context

- The context is the set of relations (tables) whose operations may cause the constraint to be violated.
- These operations can include:
  - Inserting a tuple (row)
  - Deleting a tuple
  - Updating one or more attributes within a tuple

# 2.3 Impact Analysis Table (1)

- Impact analysis identifies when an integrity constraint needs to be checked, i.e., which operations could potentially violate it.

- The impact analysis result is represented by a two-dimensional table called the *impact analysis table*.

# 2.3 Impact Analysis Table (2)

- **The impact analysis table of a constraint**

| Constraint i | Insert | Delete | Update |
|---|---|---|---|
| Relation 1 | | | |
| …… | | | |
| Relation n | | | |

How to fill each cell:

+ : perform the operation that may violate the IC

- : perform the operation that cannot violate the IC

+(X) : may violate the IC when updating on attribute X

-(*) : no violation of the IC because updating is not permitted

# 2.3 Impact Analysis Table (3)

**Some regulations:**

- It is not permitted to modify the value of a **key** attribute (i.e., an attribute that is part of the primary key of a relation).
- The impact of **insertion and deletion** operations is considered at the tuple (row) level, while the impact of **update** operations is considered at the attribute level within each tuple.
- Before determining whether an operation could violate an integrity constraint, the database must **already satisfy all integrity constraints**.

# 3. Classification of ICs

3.1 ICs with a Single-Relation Context

3.2 ICs with a Multi-Relation Context

3.3 Functional Dependency

# 3.1 Integrity Constraints with a Single-Relation Context

3.1.1 Domain Constraints

3.1.2 Inter-Attribute Constraints

3.1.3 Inter-Tuple Constraints

# Academic management database schema

**HOCVIEN** (MAHV, HO, TEN, NGSINH, GIOITINH, NOISINH, MALOP)

**LOP** (MALOP, TENLOP, TRGLOP, SISO, MAGVCN)

**KHOA** (MAKHOA, TENKHOA, NGTLAP, TRGKHOA)

**MONHOC** (MAMH, TENMH, TCLT, TCTH, MAKHOA)

**DIEUKIEN** (MAMH, MAMH_TRUOC)

**GIAOVIEN** (MAGV, HOTEN, HOCVI, HOCHAM, GIOITINH, NGSINH, NGVL, HESO, MUCLUONG, MAKHOA)

**GIANGDAY** (MALOP, MAMH, MAGV, HOCKY, NAM, TUNGAY, DENNGAY)

**KETQUATHI** (MAHV, MAMH, LANTHI, NGTHI, DIEM, KQUA)

# 3.1.1 Domain Constraints

- **A domain constraint defines the set of permissible values that an attribute can take.**
- **R1:** Student's gender is only ‘Nam’ or ‘Nữ’.
  - Content:

    $\forall$ hv $\in$ HOCVIEN: hv.Gioitinh $\in$ {‘Nam’, ‘Nữ’}
  - Context: HOCVIEN
  - Impact analysis table:

| R1 | Insert | Delete | Update |
|---|---|---|---|
| HOCVIEN | + | - | +(Lop) |

# 3.1.2 Inter-Attribute Constraints

- **Inter-attribute constraint is a constraint between attributes on a tuple of a relation.**
- **R2:** The start date (TUNGAY) of teaching a course for a class is always less than the end date (DENNGAY).
  - Content:

    $\forall$ gd $\in$ GIANGDAY: gd.TUNGAY < gd.DENNGAY
  - Context: GIANGDAY
  - Impact analysis table:

| R2 | Insert | Delete | Update |
|---|---|---|---|
| GIANGDAY | + | - | +(Tungay, Denngay) |

# 3.1.3 Inter-Tuple Constraints (1)

- **Inter-tuple constraint is a constraint between multiple tuples on the same relation (can involve multiple attributes).**
- **R3:** All students must have a unique student ID.
  - Content:

    $\forall$ h1,h2 $\in$ HOCVIEN: If h1$\neq$h2 then h1.Mahv$\neq$h2.Mahv
  - Context: HOCVIEN
  - Impact analysis table:

| R3 | Insert | Delete | Update |
|---|---|---|---|
| HOCVIEN | + | - | -(*) |

# 3.1.3 Inter-Tuple Constraints (2)

- **R4:** Teachers with the same degree and salary coefficient will have the same salary.
  - Content:

    $\forall gv_1, gv_2 \in$ GIAOVIEN:

    If $(gv_1.Hocvi = gv_2.Hocvi) \wedge (gv_1.Heso = gv_2.Heso)$

    then $gv_1.Mucluong = gv_2.Mucluong$
  - Context: GIAOVIEN
  - Impact analysis table:

| R4 | Insert | Delete | Update |
|---|---|---|---|
| GIAOVIEN | + | - | +(Hocvi, Heso, Mucluong) |

# 3.2 Integrity Constraints with a Multi-Relation Context

3.2.1 Referential Constraints

3.2.2 Inter-Attribute Constraints

3.2.3 Constraints Involving Derived Attributes

3.2.4 Constraints Caused by Cycles in the Relational Schema

# 3.2.1 Reference Constraint (1)

- **A referential constraint specifies that the value of certain attributes in one relation R must depend on the existence of corresponding attribute values in another relation S.**
  - The set of attributes in R forms a foreign key.
  - The set of attributes in S forms a primary key.
  - Referential constraints are also called *existence dependencies* or *foreign key constraints*.

# 3.2.1 Referential Constraint (2)

- **R5:** Students taking an exam for a subject must have that subject on the list of subjects.
  - Content:

    $\forall k \in$ KETQUATHI, $\exists m \in$ MONHOC: k.Mamh = m.Mamh

    **Or:** KETQUATHI[Mamh] $\subseteq$ MONHOC[Mamh]
  - Context: KETQUATHI, MONHOC
  - Impact analysis table:

| R5 | Insert | Delete | Update |
|---|---|---|---|
| KETQUATHI | + | - | -(*) |
| MONHOC | - | + | -(*) |

# 3.2.2 Inter-Attribute Constraints (1)

- **This is a constraint between the values of multiple attributes on one or more different relations.**
- **R6:** The date a teacher teaches a subject must be greater than or equal to the date the teacher starts work.
  - Content: $\forall$ gd $\in$ GIANGDAY, $\forall$ gv $\in$ GIAOVIEN:

    If gd.Magv = gv.Magv then gd.TUNGAY $\geq$ gv.NGVL
  - Context: GIANGDAY, GIAOVIEN
  - Impact analysis table:

| R6 | Insert | Delete | Update |
|---|---|---|---|
| GIANGDAY | + | - | +(Tungay) |
| GIAOVIEN | - | - | +(Ngvl) |

Requirement: Foreign keys always persist.

# 3.2.2 Inter-Attribute Constraints (2)

- **R7:** The exam date of a subject must be greater than the end date of that subject.
  - Content:

$\forall$ kq $\in$ KETQUATHI, $\forall$ hv $\in$ HOCVIEN, $\forall$ gd $\in$ GIANGDAY:

If (kq.Mamh=gd.Mamh)

$\wedge$ (kq.Mahv=hv.Mahv)

$\wedge$ (hv.Malop=gd.Malop)

then kq.Ngthi > gd.Denngay

Tips: Convert multi-relation context into single-relation context, using Join (put all data on 1 row).

  - Context: KETQUATHI, HOCVIEN, GIANGDAY

# 3.2.2 Inter-Attribute Constraints (3)

- Impact analysis table:

| R7 | Insert | Delete | Update |
|---|---|---|---|
| KETQUATHI | + | - | +(Ngthi) |
| HOCVIEN | - | - | +(Malop) |
| GIANGDAY | - | - | +(Denngay) |

Requirement:
Foreign keys
always persist.

# 3.2.3 Constraints Involving Derived Attributes (1)

- **This is a constraint between attribute values or tuples across different relations, through a derived attribute.**
  - A derived attribute is one that is computed from other attributes or tuples.

# 3.2.3 Constraints Involving Derived Attributes (2)

- **For example:**

  SANPHAM(<u>Masp</u>, Tensp, Nuocsx, Gia)
  KHACHHANG(<u>Makh</u>, Hoten, Doanhso)
  HOADON(<u>Sohd</u>, Nghd, Makh, Trigia)
  CTHD(<u>Sohd, Masp</u>, Soluong, Gia)

→ The value of an invoice (Trigia) is equal to the total amount of the invoice details of that invoice (Gia).

→ A customer's sales (Doanhso) is equal to the total value of the invoices that the customer has purchased (Trigia).

# 3.2.3 Constraints Involving Derived Attributes (3)

**For example:** *A customer's sales is equal to the total value of the invoices that the customer has purchased.*

- Content:

  $\forall kh \in KHACHHANG:$

  $$kh.Doanhso = \sum_{(hd \in HOADON:\ hd.Makh=kh.Makh)} (hd.Trigia)$$

- Context: KHACHHANG, HOADON
- Impact analysis table:

| | Insert | Delete | Update |
|---|---|---|---|
| KHACHHANG | + | - | +(Doanhso) |
| HOADON | + | + | +(Makh, Trigia) |

Requirement: Foreign keys always persist.

But when insert new: Doanhso = 0.

# 3.2.3 Constraints Involving Derived Attributes (4)

- **R8:** Class size is the number of students in that class.
  - Content:

    $\forall l \in LOP$:

    $l.Siso = Count_{(hv \in HOCVIEN:\ hv.Malop = l.Malop)}(*)$
  - Context: LOP, HOCVIEN
  - Impact analysis table:

| R8 | Insert | Delete | Update |
|---|---|---|---|
| LOP | + | - | +(Siso) |
| HOCVIEN | + | + | +(Malop) |

Requirement: Foreign keys always persist.

But when insert new: class size = 0.

# 3.2.4 Constraints Caused by Cycles in the Relational Schema (1)

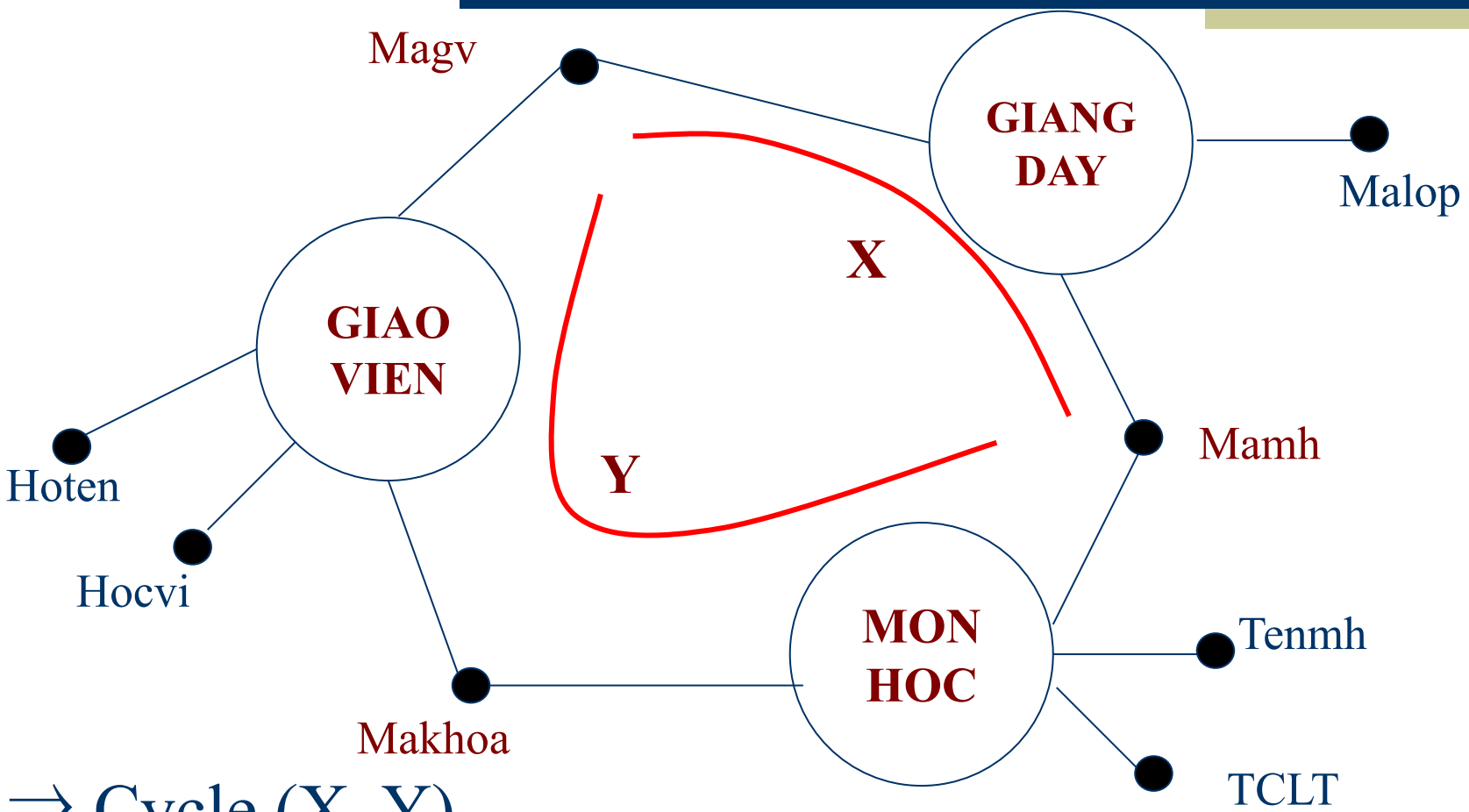
To representing a relational schema as a graph:

- Relations are represented by large hollow circles.
- Attributes are represented by small filled circles.
- Connect the relation and its attributes by an edge linking the large and small circles.

If the graph contains a closed path,

$\Rightarrow$ The schema shows the presence of a cycle.

# 3.2.4 Constraints Caused by Cycles in the Relational Schema (2)




⇒ Cycle (X, Y)

# 3.2.4 Constraints Caused by Cycles in the Relational Schema (3)

- Cycle (X, Y):
  - X = GIANGDAY[Magv, Mamh]
  - Y = (GIAOVIEN $\overset{\text{Makhoa}}{\bowtie}$ MONHOC) [Magv, Mamh]
- **Meaning:**
  - X: teachers and the subjects assigned to that teacher to teach.
  - Y: teachers and subjects managed by that teacher's faculty.

# 3.2.4 Constraints Caused by Cycles in the Relational Schema (4)

- **Types of constraints related to a cycle (X, Y):**
  - **Constraint 1:** A teacher can only be assigned to teach subjects managed by that teacher's faculty: $X \subseteq Y$
  - **Constraint 2:** A teacher must be assigned to teach all subjects managed by that teacher's faculty: $X = Y$
  - **Constraint 3:** Teachers can be assigned to teach any subject: X and Y are arbitrary

# 3.2.4 Constraints Caused by Cycles in the Relational Schema (5)

- **R9** : A teacher can only be assigned to teach subjects managed by that teacher's faculty: **X⊆Y**

| R9 | Insert | Delete | Update |
|---|---|---|---|
| MONHOC | - | - | +(Makhoa) |
| GIAOVIEN | - | - | +(Makhoa) |
| GIANGDAY | + | - | +(Magv) |

Requirement: Foreign keys always persist.

# 3.3 Functional Dependency (1)

- A functional dependency is a special form of IC.
- Every relation has at least one functional dependency.
- Given a relation Q(A, B, C), the functional dependency "A determines B", denoted $A \rightarrow B$, is defined as:

  $\forall q_1, q_2 \in Q$: If $q_1.A = q_2.A$, then $q_1.B = q_2.B$

# 3.3 Functional Dependency (2)

Example:

- A key constraint is also a functional dependency:

  Mamh → Tenmh, TCLT, TCTH, Makhoa

- (R4) *Teachers with the same degree and salary coefficient will have the same salary.*

  This constraint can be expressed as a functional dependency: Hocvi, Heso → Mucluong

# 3.3 Functional Dependency (3)

**Trivial Functional Dependency:**

- A functional dependency $A \rightarrow B$ is called *trivial* if $B \subseteq A$.
  - That is, the set of attributes A already includes B.

- Example: Let A = (MaHV, HoTen), B = (HoTen).
  - Since HoTen $\subseteq$ A, the dependency $A \rightarrow B$ is a trivial functional dependency.

# 3.3 Functional Dependency (4)

**Prime (Minimal) Functional Dependency:**

- $A \rightarrow B$ is called *prime* (minimal) if $\neg \exists A' \subset A, A' \neq A : A' \rightarrow B$.
  - In other words, A is the smallest **subset** of attributes that determines B.

- Example: Let A = (MaHV, HoTen), B = (NgSinh), $A \rightarrow B$.
  - We have a smaller subset A' = (MaHV) satisfying $A' \rightarrow B$
    - So $A \rightarrow B$ is not a prime functional dependency.
    - Instead, $A' \rightarrow B$ is the prime functional dependency.

# 4. Summary Impact Analysis Table (1)

- **The summary table** of m constraints on n context relations:

| | **$Rel_1$** | | | **$Rel_2$** | | | **…** | **$Rel_n$** | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | I | D | U | I | D | U | … | I | D | U |
| R1 | | | | | | | | | | |
| R2 | | | | | | | | | | |
| … | | | | | | | | | | |
| Rm | | | | | | | | | | |

# 4. Summary Impact Analysis Table (2)

| | HOCVIEN | | | GIAOVIEN | | | LOP | | | MONHOC | | | GIANGDAY | | | KETQUA THI | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | I | D | U | I | D | U | I | D | U | I | D | U | I | D | U | I | D | U |
| **R1** | + | - | + | | | | | | | | | | | | | | | |
| **R2** | | | | | | | | | | | | | + | - | + | | | |
| **R3** | + | - | -* | | | | | | | | | | | | | | | |
| **R4** | | | | + | - | + | | | | | | | | | | | | |
| **R5** | | | | | | | | | | - | + | -* | | | | + | - | -* |
| **R6** | | | | - | - | + | | | | | | | + | - | + | | | |
| **R7** | - | - | + | | | | | | | | | | - | - | + | + | - | + |
| **R8** | + | + | + | | | | + | - | + | | | | | | | | | |
| **R9** | | | | - | - | + | | | | - | - | + | + | - | + | | | |